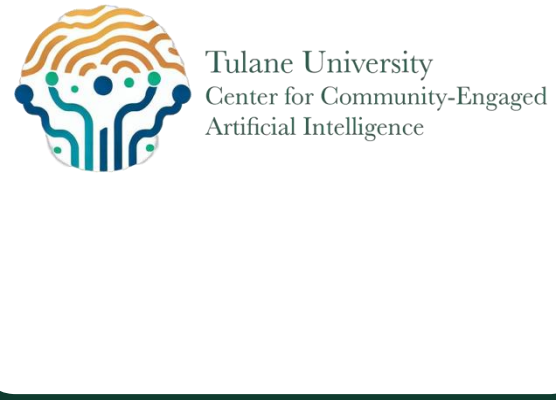
IARSLCE 2026

Sun, Oct 11 · 9:00 – 11:30

IARSLCE 2026 · PRE-CONFERENCE WORKSHOP

Participatory AI for Community Engagement



Nicholas Mattei & Saad Hassan

Tulane Center for Community-Engaged Artificial Intelligence

Sunday, October 11, 2026 · 9:00 – 11:30 am
Monroe Hall 152, Loyola University, New Orleans

Welcome | AI | Design | Reflect | Share | Break | Breakouts | Report | Debrief

Today's plan

9:00 – 9:10 · 10 min

9:00	Welcome and introductions	Nick & Saad
9:10	What is AI, and where are we?	Nick
9:25	Designing with, not for	Saad
9:40	**Reflect: your one problem**	Everyone
9:55	**Share-outs**	Everyone
10:10	Break	We cluster your ideas
10:25	**Breakout groups: three rounds**	Two groups
11:10	**Report back**	Each group
11:20	**Debrief and next steps**	Everyone

Slides, readings, and links

saadh.info/pace2

Introductions

9:00 – 9:10 · 10 min



nickmattei.net

Nicholas Mattei

Associate Professor of Computer Science, Tulane University
Co-Director, Tulane Center for Community-Engaged AI

- AI, machine learning, and decision making with real-world stakeholders
- Community projects across New Orleans: civic tech, courts, utilities
- Co-author, Computing and Technology Ethics: Engaging Through Science Fiction (MIT Press)

Chair, ACM SIGAI | NSF CAREER | Prev. NASA · IBM · UNSW

Introductions

9:00 – 9:10 · 10 min



saadh.info

Saad Hassan

Assistant Professor of Computer Science, Tulane University
Director, NOLA A11y Lab · Affiliate, CEAI

- Human–AI systems that make communication, information, and public services accessible
- Sign language and captioning technologies with Deaf and hard-of-hearing communities
- Civic AI and accessibility dashboards with community organizations

PhD, RIT | Visiting: Google & Meta | CHI · ASSETS · NeurIPS

Let's get to know you

9:00 – 9:10 · 10 min

1

Your name

and how you'd like to be called

2

Your affiliation

institution, office, or organization

3

One community partner or project

you're working with right now

About 30 seconds each

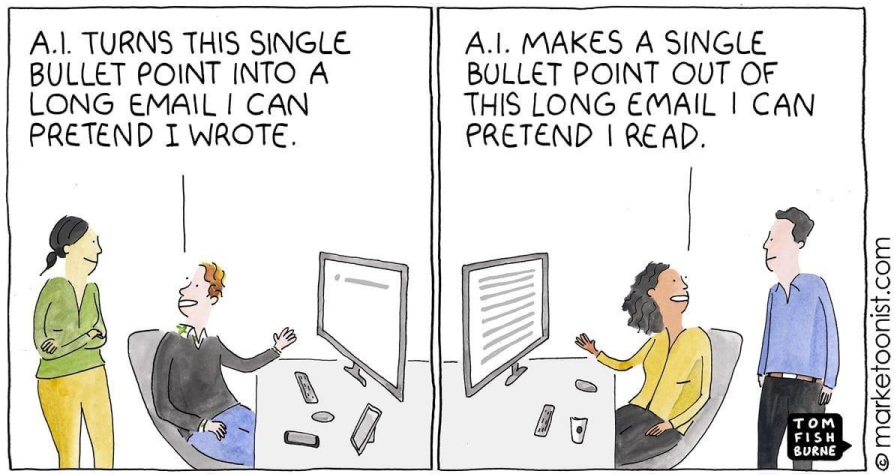
The pre-workshop survey covers the rest, so keep it short.

9:10 – 9:25 · 15 min

PART 1

What is AI, and where are we?

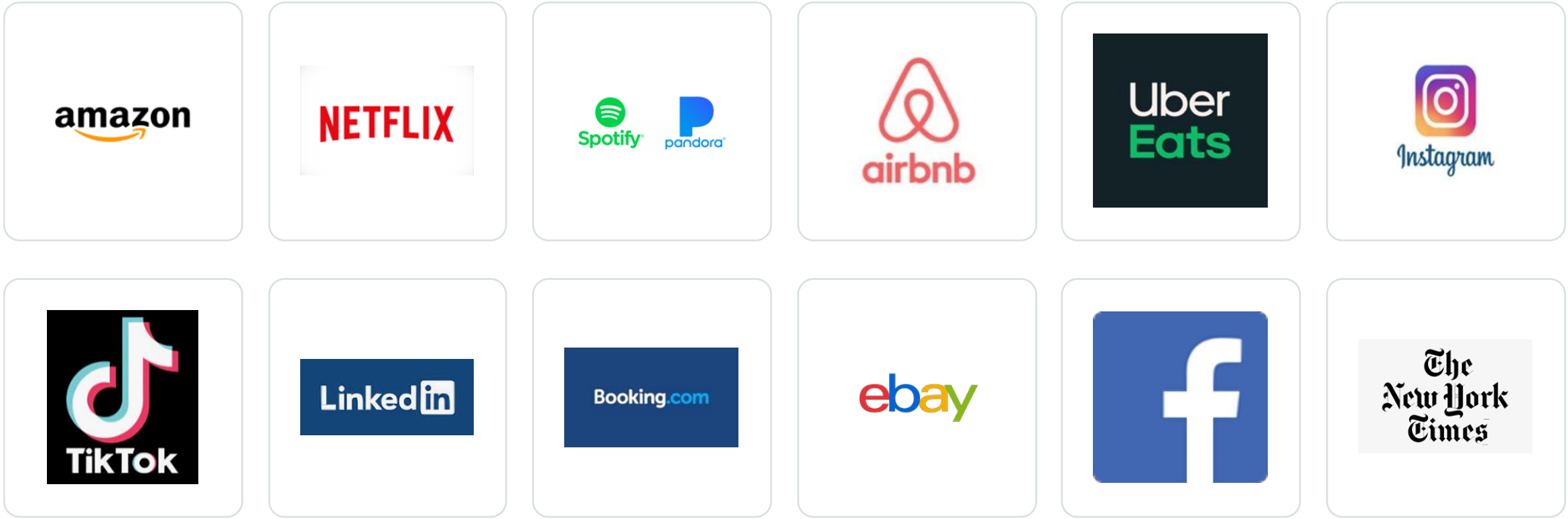
Nicholas Mattei



AI is already everywhere

9:10 – 9:25 · 15 min

Feeds, search, prices, matches, rides, loans: most of it has run on AI for years.



Everyone is a programmer now

9:10 – 9:25 · 15 min

THE GOAL

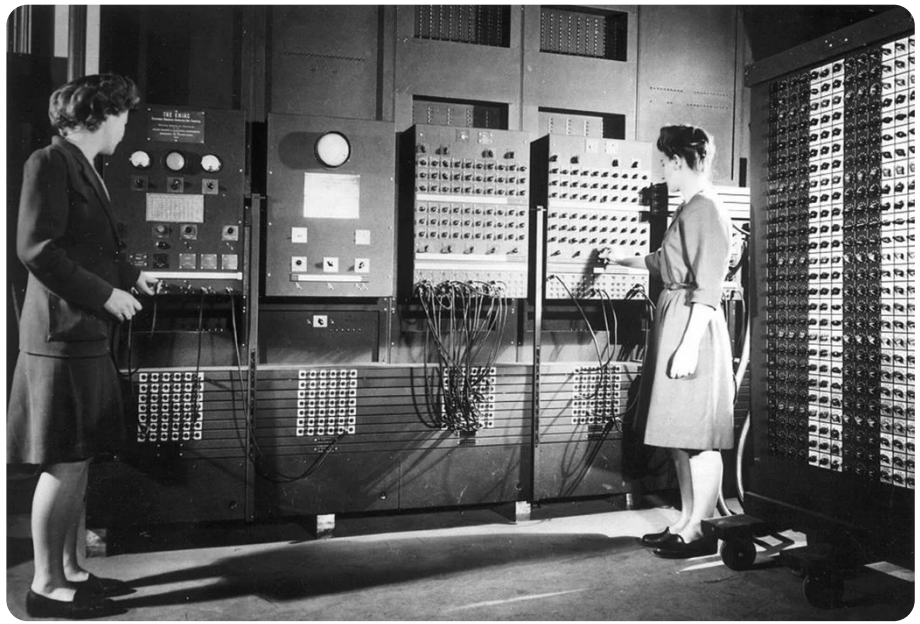
Make intent clear to computers!

Wires › Punch cards › Languages › No-code › **Prompt as code**

CS folks should **embrace this!**

The magic of CS is that we get to make the machine do what we want. Now everyone can!

Secondary argument: we've been in the bot age for a while (search, etc. is the product). People never produced this… (well, in the 90s maybe).





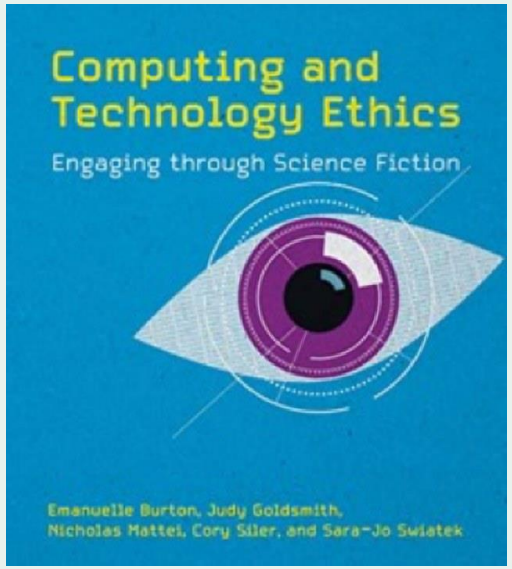
READ MORE

[The End of Programming as We Know It](#) (O'Reilly)

Teaching computing and technology ethics

9:10 – 9:25 · 15 min

Engaging through science fiction



AVAILABLE NOW · MIT PRESS

CHAPTERS

- Ethical Frameworks
- Managing Knowledge
- Privacy and Personhood
- Technology and Society
- Professional Ethics

ANTHOLOGY

Short stories from acclaimed writers including Ken Liu, Elizabeth Bear, Paolo Bacigalupi, and T. C. Boyle animate ethical challenges in computing technology.







GET THE BOOK

rebrand.ly/SciFi-Book

Let’s step back: what is technology?

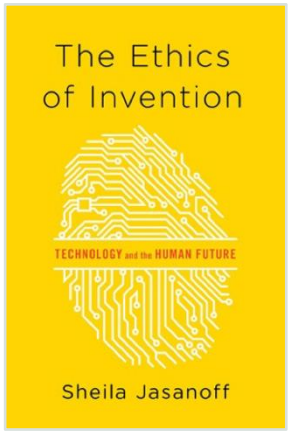
9:10 – 9:25 · 15 min

1. **AI and technology is not a single thing or only objects.**
   It does not have agency, it is not a person, and by many philosophical/religious arguments cannot be “***ethical***.”

2. **Focusing only on specific objects or artifacts: a camera, a traffic light, a gun, an app, a bike…**
   How technology affects society is an insufficient question of cause and effect: ***does a gun do bad things?***

3. **Technology and society are bound together in an inseparable way: our choices about which technologies to develop, or not to develop, often grow out of the society in which we live.**
   And those technologies, in turn, change our social settings and the subsequent technologies that are produced in the context in which we live.

Consider the stoplight…



Sheila Jasanoff
The Ethics of Invention



Consider the stoplight

9:10 – 9:25 · 15 min

INNOVATION

A Brief History of the Stoplight

How a bright idea shaped our cities and gave the go-ahead to our love affair with the car

Megan Kate Nelson

Author, *Saving Yellowstone: Exploration and Preservation in Reconstruction America*

May 2018

An additional complaint that gained traction was the device's unfortunate impact on civility. Long before today's epidemic of road rage, critics warned that drivers had surrendered some of their humanity; they didn't have to acknowledge each other or pedestrians at intersections, but rather just stare at the light and wait for it to change. As



THE FIRST SIGNAL

London, 1868. A gas-lit semaphore outside the Houses of Parliament.

the Houses of Parliament.[8] The design combined three semaphore arms with red and green gas lamps for night-time use, on a pillar, operated by a police constable. The gas lantern was manually turned by a traffic police officer with a lever at its base so that the appropriate light faced traffic.[9] The signal was 22 feet (6.7 m) high. The light was called the semaphore and had arms that would extend horizontally that commanded drivers to "Stop" and then the arms would lower to a 45 degrees angle to tell drivers to proceed with "Caution". At night a red light would command "Stop" and a green light would mean use "Caution".[4] Although it was said to be successful at controlling traffic, its operational life was brief. It exploded on 2 January 1869 as a result of a leak in one of the gas lines underneath the pavement[10] and injured the policeman who was operating it.[11]

Sources: Smithsonian Magazine · Wikipedia, History of traffic lights

Wait, what is AI?

9:10 – 9:25 · 15 min

ALAN TURING, 1950

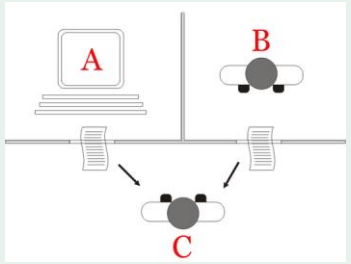
“I propose to consider the question, ‘Can machines think?’ … I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.”



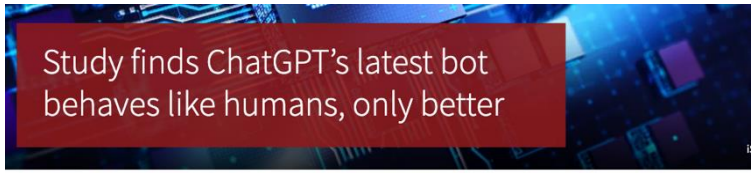


THE IMITATION GAME

“… a man (A), a woman (B) and an interrogator (C) … The object of the game for the interrogator is to determine which of the other two is the man and which is the woman.”



TODAY



The most recent version of ChatGPT passes a rigorous Turing test, diverging from average human behavior chiefly to be more cooperative.

Passes a rigorous Turing test. Scan for the study.

Tessler’s Theorem: *“Intelligence is what machines haven’t done yet.”*

I.e., “If machines can do it, it isn’t intelligence.”

AI for AI researchers

9:10 – 9:25 · 15 min



"'Artificial intelligence' is when a machine mimics 'cognitive functions' that humans associate with other human minds such as 'learning' and 'problem solving.'"

AGENT-BASED APPROACH

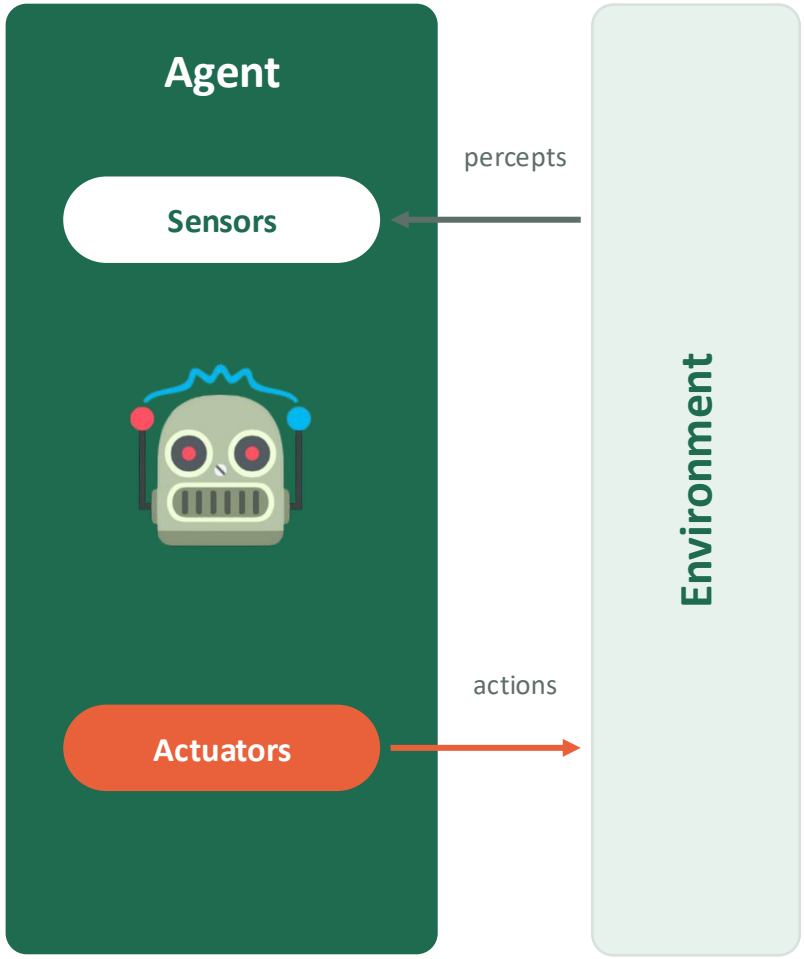
"... rational agents that act so as to achieve the best outcome, or, when there is uncertainty, the best expected outcome."

TRADITIONAL GOALS

Reasoning, planning, and knowledge representation

Machine learning and natural language processing

Perception (vision) and embodiment (robotics)



Text-to-text systems

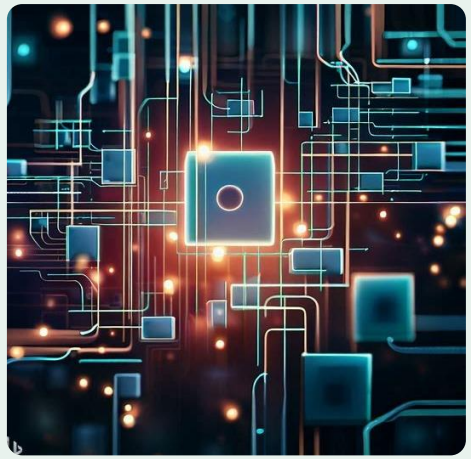
9:10 – 9:25 · 15 min



01

The data

Scrape LOTS of TEXT from the internet.



02

The model

Create a giant statistical model that relates how WORDS appear with other WORDS.

give me the Gettysburg address

03

The prompt

You type a request.

Four score and seven years ago our forefathers brought…

04

The output

The model continues the text, one word at a time.

Given the words so far, keep predicting

9:10 – 9:25 · 15 min

The best thing about AI is its ability to,

The best thing about AI is its ability to **create**,

The best thing about AI is its ability to create **worlds**,

The best thing about AI is its ability to create worlds **that**,

The best thing about AI is its ability to create worlds that **are**,

The best thing about AI is its ability to create worlds that are **both**,

The best thing about AI is its ability to create worlds that are both **exciting**,

Source: Stephen Wolfram, “What Is ChatGPT Doing … and Why Does It Work?” (2023)

General idea: predict the next word

9:10 – 9:25 · 15 min



The best thing about AI is its ability to …

MOST LIKELY NEXT WORDS



Values from Stephen Wolfram, “What Is ChatGPT Doing … and Why Does It Work?” (2023)

Three ways to tell a computer what to do

9:10 – 9:25 · 15 min

1

Traditional programming

The computer does exactly what you tell it, whether you meant it or not.

Rules written by people

2

Machine learning

Learns patterns from data to predict, classify, and recommend.

Rules learned from data

3

Generative AI

Trained broadly to do many things. Wonderful new behavior, but hard to predict exactly.

Text, images, code on demand

From autocomplete to agents

9:10 – 9:25 · 15 min

01

Large language model

Guess the next word

Words in, word out. Trained on huge amounts of text.

›

02

Chatbot

Hold a conversation

Tuned with human feedback to follow instructions.

›

03

Reasoning model

Work through a problem

Writes hidden steps before it answers.

›

04

Agentic system

Act in the world

Searches the web, runs code, sends email, edits files.

Each step adds capability, and each step adds new ways to fail.

The good, the bad, and the ugl-AI

9:10 – 9:25 · 15 min



- Protein structure prediction (AlphaFold)
- Real progress on language tasks



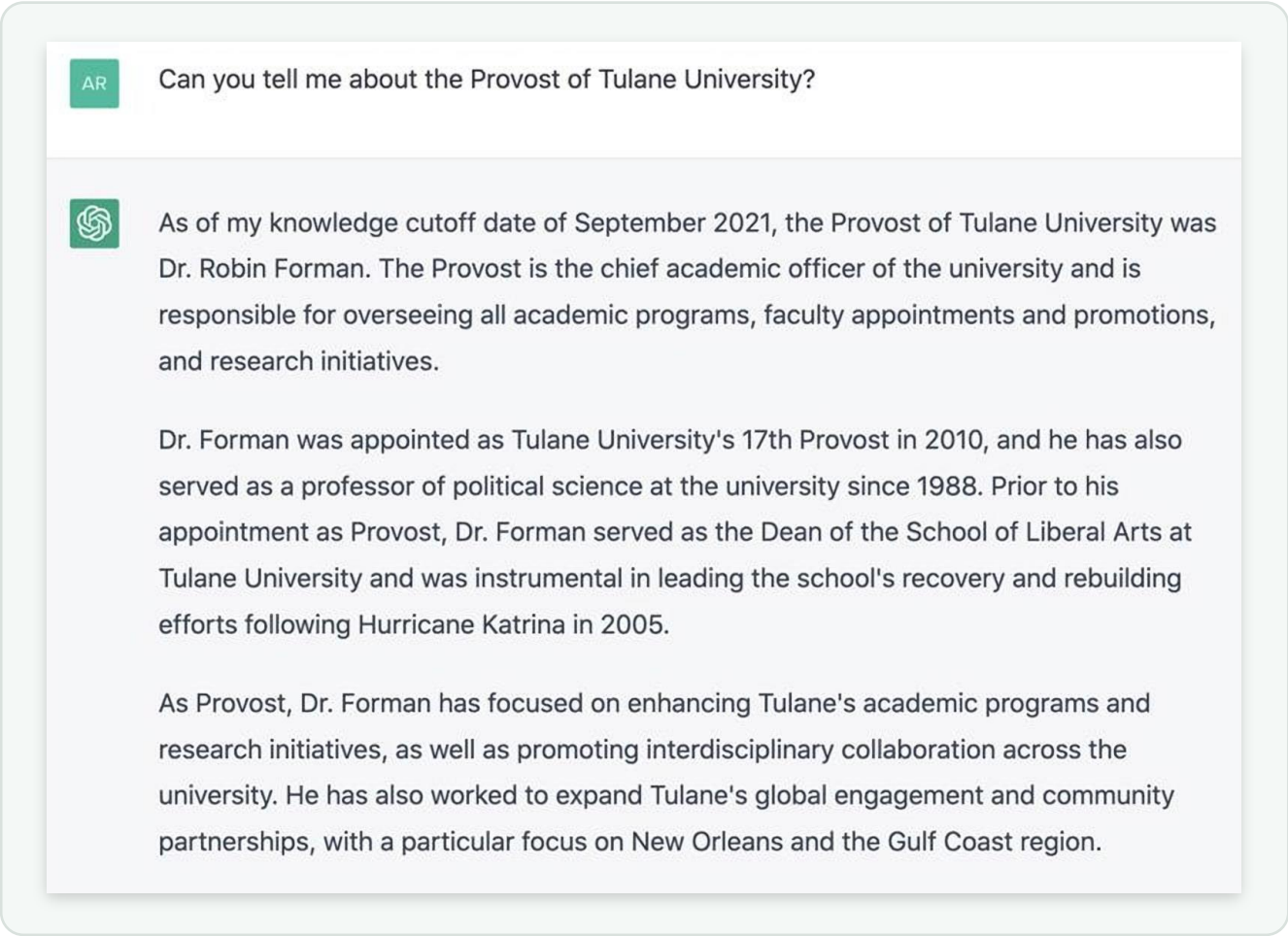
- Rising electricity costs
- Copyright and intellectual property



- Question answering, but it makes things up
- Study aid, but it can block learning
- Companionship, but people can spiral

Fluent, confident, and wrong

9:10 – 9:25 · 15 min



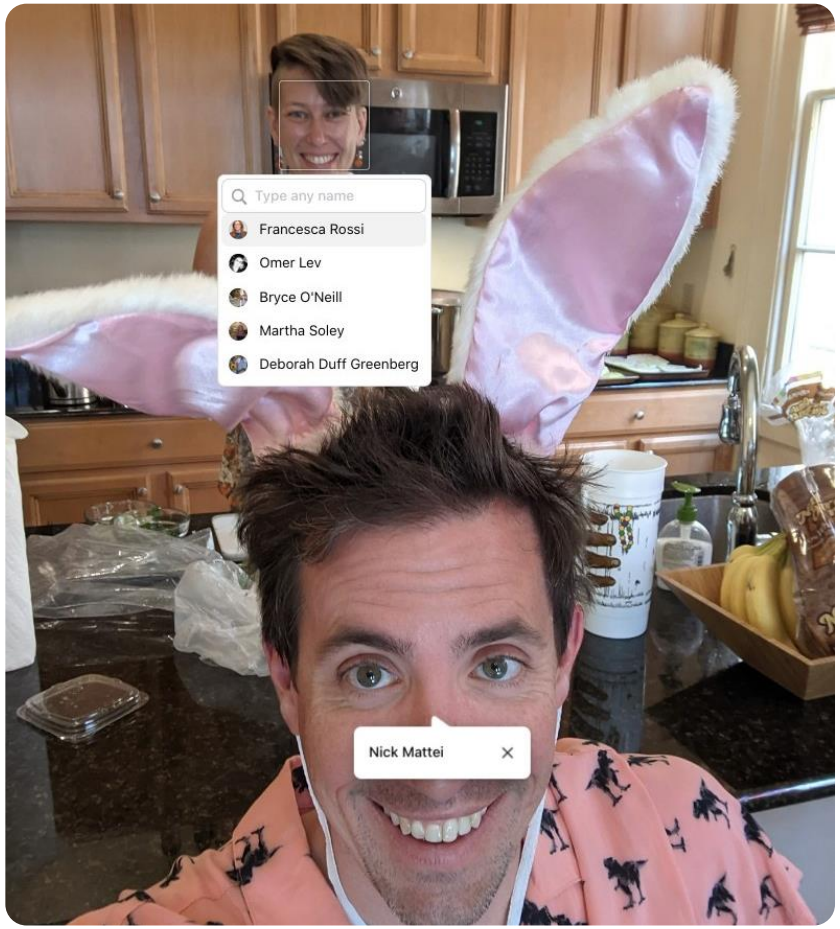
Ask a chatbot about Tulane's provost and you get a detailed, polished biography, with details that are made up.

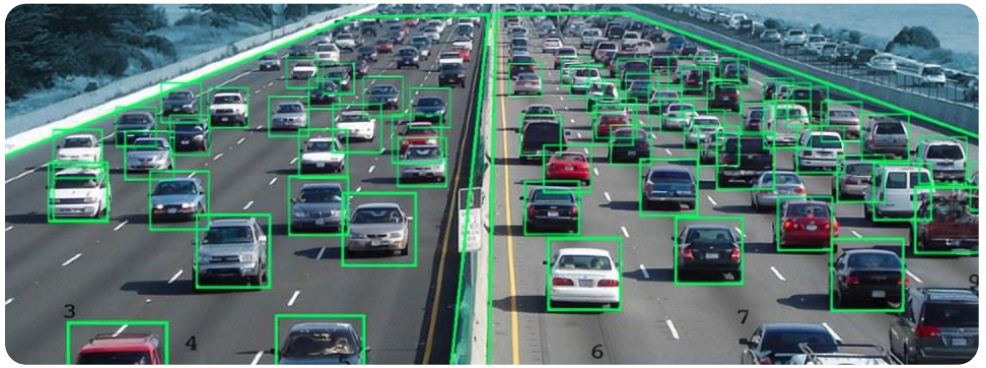
Often called “hallucination.” Nick avoids the word: it anthropomorphizes a random number generator.

Check every output. Test, test, and test again.

What’s at stake?

9:10 – 9:25 · 15 min

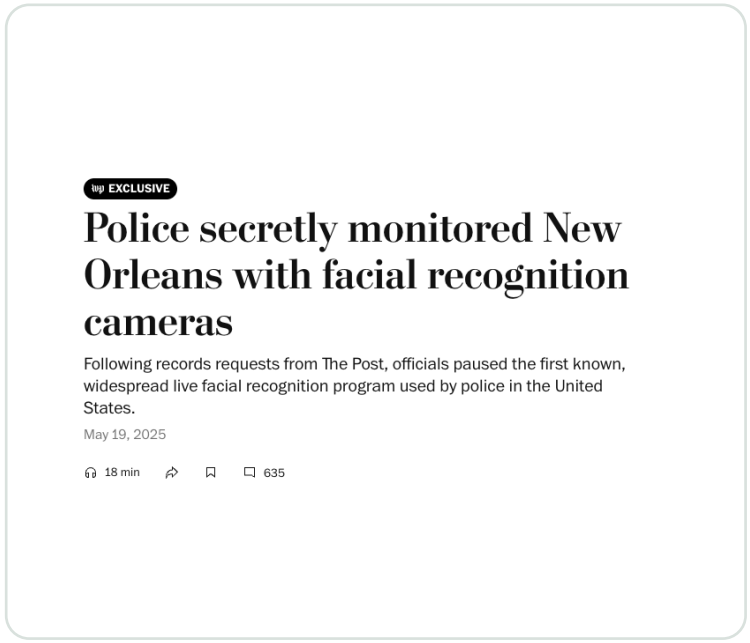


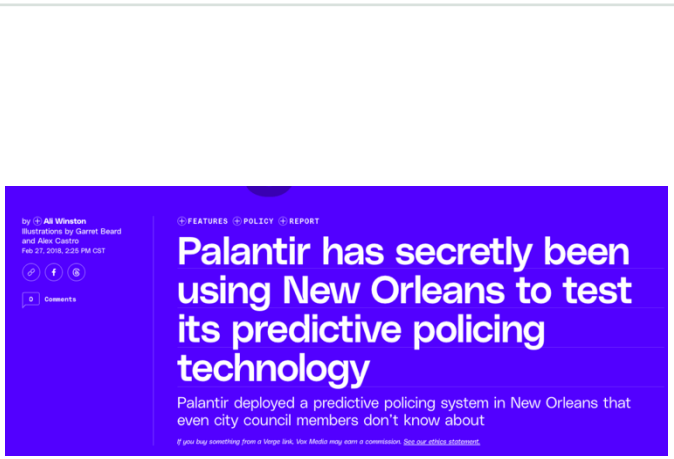


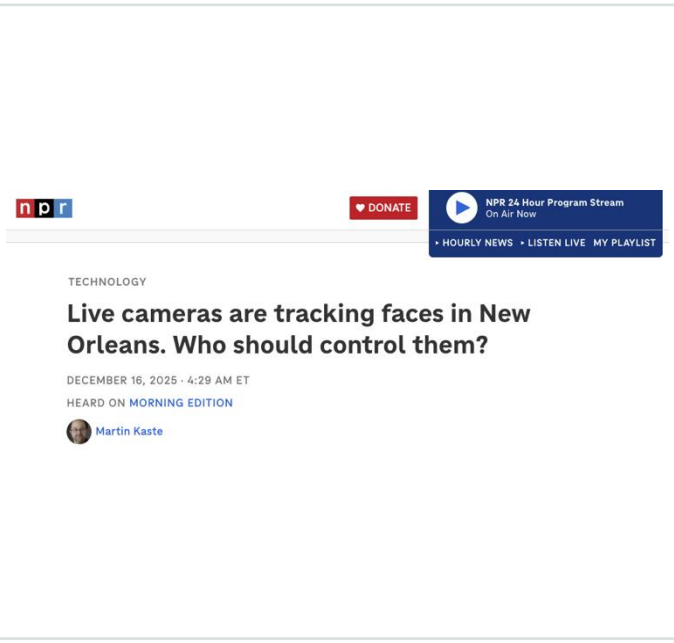


What's at stake, right here in New Orleans

9:10 – 9:25 · 15 min



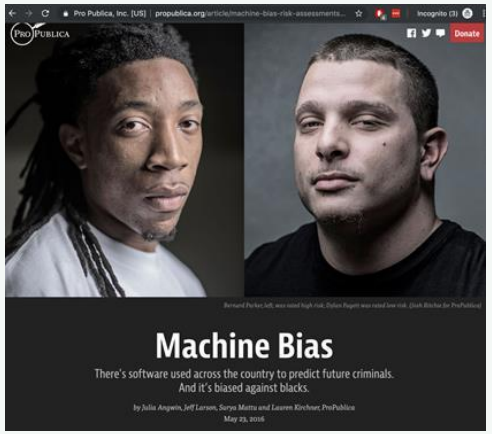


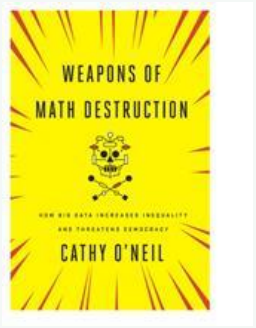


Facial recognition and predictive policing were tested on New Orleans residents, often without public knowledge.

Automated decisions of consequence

9:10 – 9:25 · 15 min

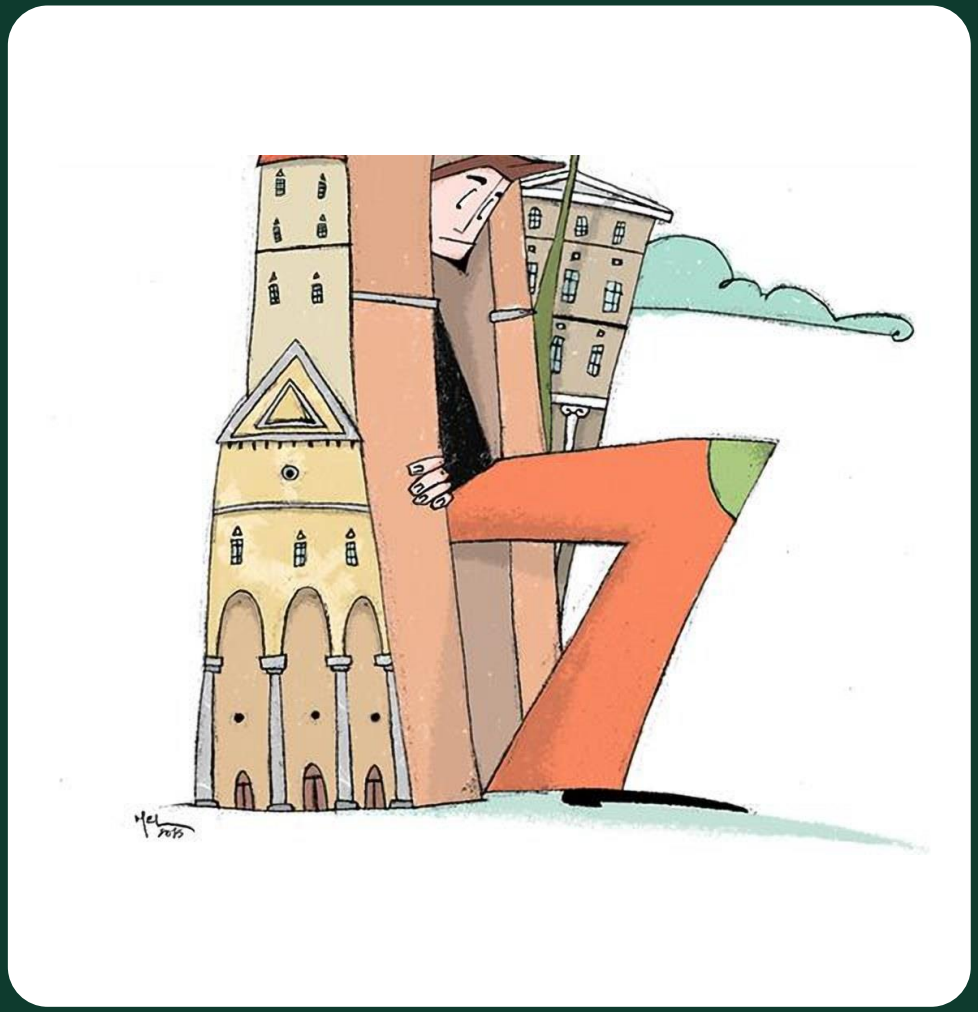




ProPublica, Machine Bias (2016)
O'Neil, Weapons of Math Destruction (2016)

Hiring	A recruiting tool learned to downgrade women's résumés and was scrapped.
Advertising	Searching Black-identifying names surfaced more ads for arrest records.
Lending	Credit models can reproduce historical patterns of exclusion.
Policing and sentencing	Risk scores shape bail and sentencing decisions.

9:10 – 9:25 · 15 min



How can academia collaborate outside the ivory tower to solve problems using AI?

Illustration: Melania Bianucci, Ludicomix

A community-led research pipeline

9:10 – 9:25 · 15 min



Community workshops

Learn from the community. Build relationships with non-profits.

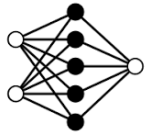
›



Service-learning capstones

Year-long student projects with partners. Working prototypes.

›



Research teams

Faculty from CS, sociology, economics, public health.

›



Joint funding

Seed funds, then shared grants with partners as co-PIs.

Tulane University
Center for Community-Engaged Artificial Intelligence

sse.tulane.edu/cs/ceai

LLMs for city government

9:10 – 9:25 · 15 min

EXAMPLE 1 with Eye on Surveillance

THE CHALLENGE

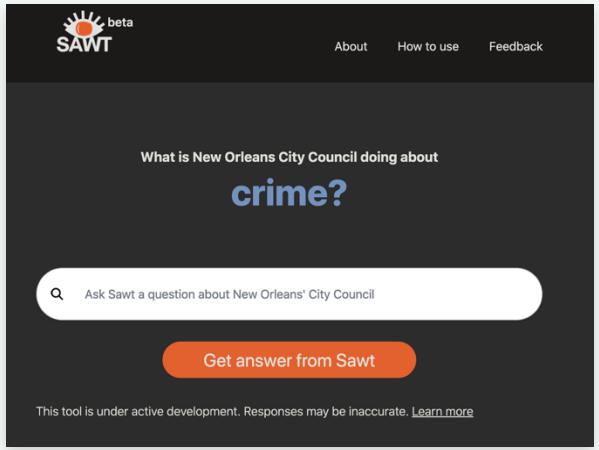
City Council meetings are hours long and hard for residents to search.

WHAT WE BUILT

SAWT: ask a question, get answers drawn from council meeting records.

LEARN MORE

sawt.us





REGGIE: searchable utility meetings

9:10 – 9:25 · 15 min

EXAMPLE 2 with Entergy

THE CHALLENGE

Public Service Commission meetings: 5 jurisdictions, 5+ hour videos every month.

WHAT WE BUILT

REGGIE, a chatbot that indexes public meetings for question answering.

IMPACT

PSC research went from hours to minutes. Best CS Project, Tulane Capstone Expo.



entergy

AI for monitoring criminal court

9:10 – 9:25 · 15 min

EXAMPLE 3 with Court Watch NOLA

THE PARTNER

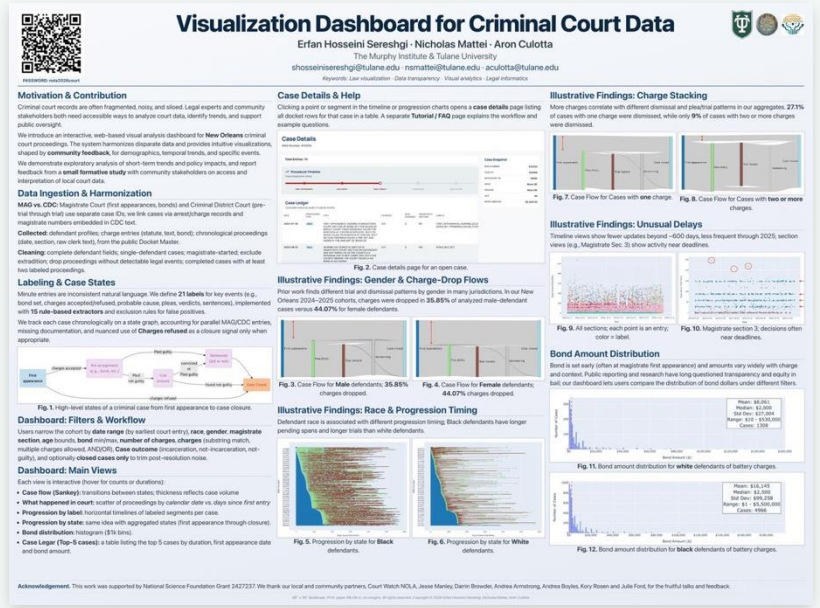
Since 2007, trained volunteers have observed Orleans Parish criminal court.

STARTED SMALL

Student-built data infrastructure for 100K+ cases over 15 years.

GREW INTO

An NSF-funded program: anomaly detection and LLM tools for defendants, victims, and attorneys.



Fairness in microloan discovery

9:10 – 9:25 · 15 min

EXAMPLE 4 with Kiva.org

THE PARTNER

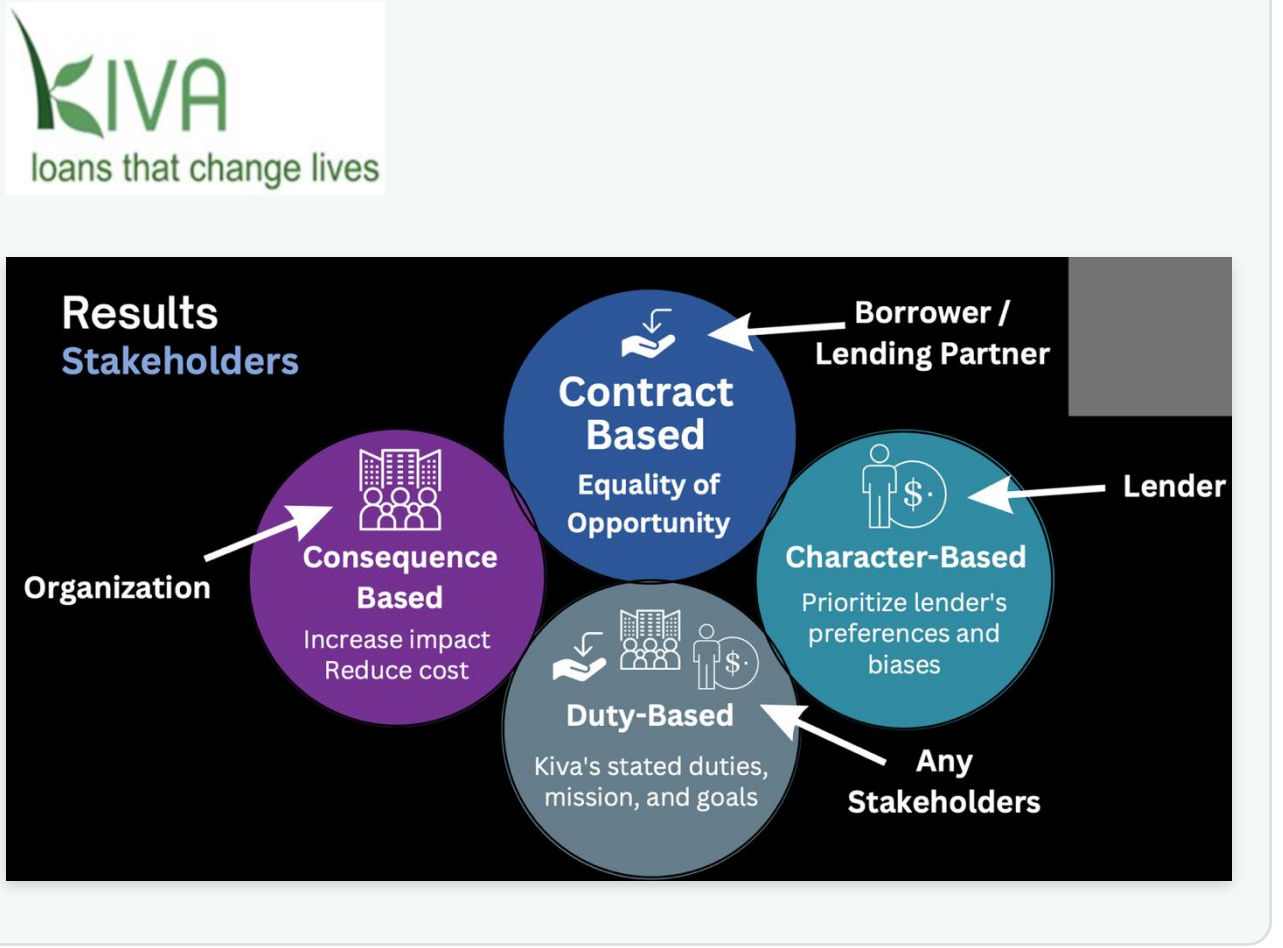
Kiva crowdfunds microloans: $1.74B to 4.3M borrowers in 77 countries (2022).

THE CHALLENGE

Lender attention is uneven, so some borrowers get overlooked.

WHAT WE STUDIED

Recommendations that balance fairness across borrowers, lenders, and Kiva's mission.



Good practices we've learned

9:10 – 9:25 · 15 min

1	**Start with conversations**	Outreach and ideation meetings, with help from your public service office.
2	**Start small**	Data prep, websites, capstone-sized projects. Early wins build trust.
3	**Share the money and the credit**	Partners as co-PIs. Budget for their time.
4	**Plan for years, not semesters**	Hand work off between student cohorts.
5	**Listen deeply**	Including about when not to use AI.

“Not a pipeline in which communities supply problems that are appropriated by academics.”

Culotta & Mattei, Service Learning for Community-Centered AI Research

9:25 – 9:40 · 15 min

PART 2

Designing with, not for

Saad Hassan

HCI, and user-centered, human-centered, and participatory design

What is HCI?

9:25 – 9:40 · 15 min

“A discipline concerned with the design, evaluation, and implementation of interactive computing systems for human use.”

ACM SIGCHI Curricula for Human-Computer Interaction, 1992

1. **Understand people**
2. **Design with them**
3. **Build interactive systems**
4. **Evaluate in real use**

DRAWS ON

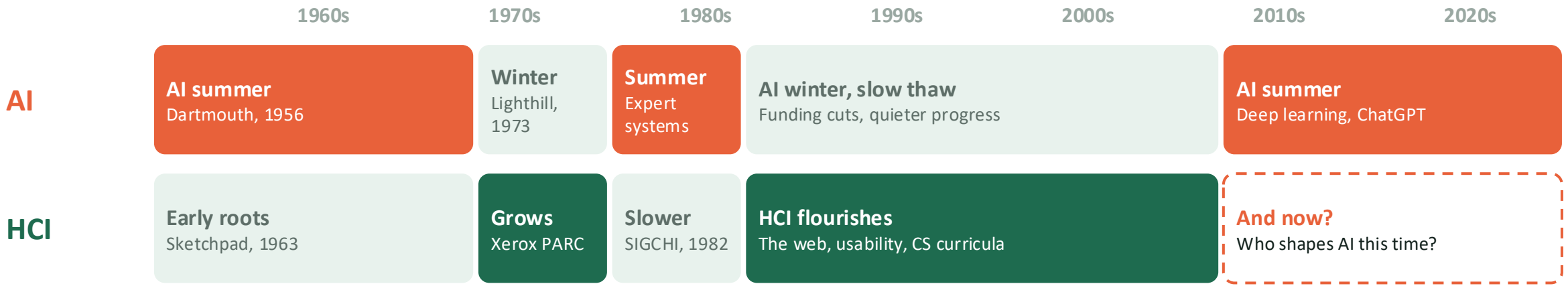
- Computer science
- Psychology and cognitive science
- Design
- Sociology and anthropology
- Disability studies

AI summers, HCI winters, and vice versa

9:25 – 9:40 · 15 min

HCI has flourished in AI winters, and moved more slowly when AI was in favor.

Grudin, "AI and HCI: Two Fields Divided by a Common Focus," AI Magazine, 2009



We are in another AI summer.

This is when HCI and participatory design matter most.

What is design?

9:25 – 9:40 · 15 min

“The transformation of existing conditions into preferred ones.”

Herbert Simon, 1969



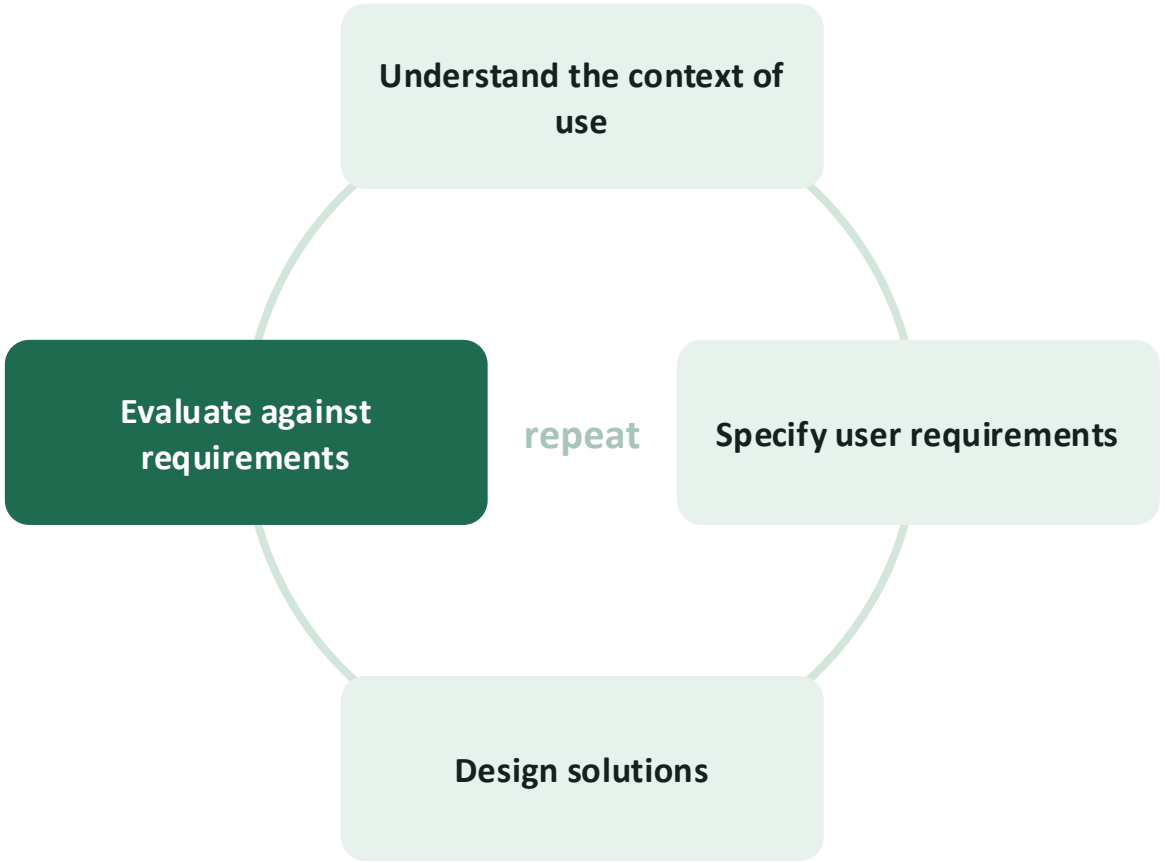
Preferred by whom?

Community problems are wicked problems: requirements conflict or are unknown, and there is no single correct answer (Rittel & Webber, 1973).

Widen the sidewalks, and traffic shifts onto the streets.

User-centered design (UCD)

9:25 – 9:40 · 15 min



Early and continual focus on users, measurement of real use, and iterative design.

Norman & Draper, 1986

The limitation

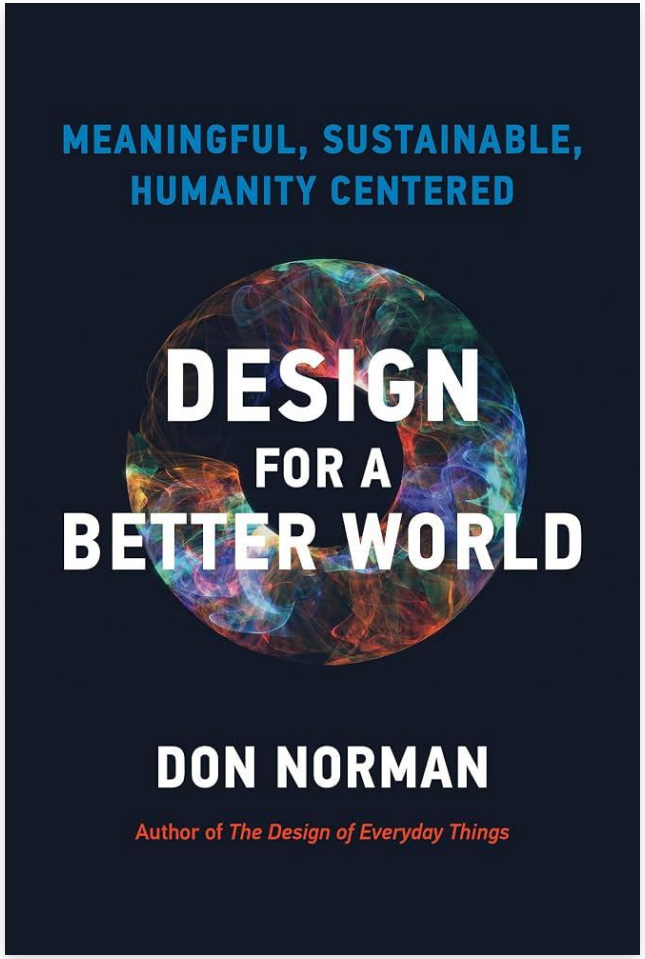
Users are studied and tested with, but designers still decide what counts as a requirement. Power stays with the design team.

Human-centered design (HCD)

9:25 – 9:40 · 15 min

Widens the lens from users to people, communities, and the systems around them.

1. **Immersion**
2. **Solve the root problem**
3. **Think in systems**
4. **Iterate, small and often**



Two warnings from the research

9:25 – 9:40 · 15 min

Is empathy enough?

Simulating disability (for example, wearing a blindfold) can lend false confidence and lead to wrong conclusions.

Bennett & Rosner, “The Promise of Empathy: Design, Disability, and Knowing the ‘Other’,” CHI 2019

Is a seat at the table enough?

Bringing people into the room does not mean they have, or see themselves as having, full status and authority.

Harrington, Erete & Piper, “Deconstructing Community-Based Collaborative Design,” CSCW 2019

Participatory design (PD)

9:25 – 9:40 · 15 min

De-centers the designer. Stakeholders share decisions in every phase.

Schuler & Namioka, 1993

Needfinding › **Problem definition** › **Developing ideas** › **Evaluation**

EXAMPLES

Gig workers co-designing a collective action platform

Patients and doctors shaping medical translation tools

Older adults designing health tracking apps

Between Scylla and Charybdis

9:25 – 9:40 · 15 min



The Odyssey (Universal Pictures, 2025)

Too few voices

"You should have included group X." A design that misses the mark.

Too many, with no stopping point

Is there an ethical point to stop adding voices and still make decisions?

Where can communities shape AI?

9:25 – 9:40 · 15 min

1

Problem framing

What should we work on?

2

Data

Whose data, collected how?

3

Design

What does it look like?

4

Evaluation

Who decides it works?

5

Governance

Who deploys and oversees it?

And the question before all of these:

Should AI be used here at all?

Sign language for DHH signers

9:25 – 9:40 · 15 min

EXAMPLE 1 **Deaf-led, with Google and DPAN**

PERSONAL ASSISTANTS

Assistants and smart speakers assume speech. We study how Deaf signers want to use them in sign language.

TEXT ENTRY · MOBILEHCI 2023

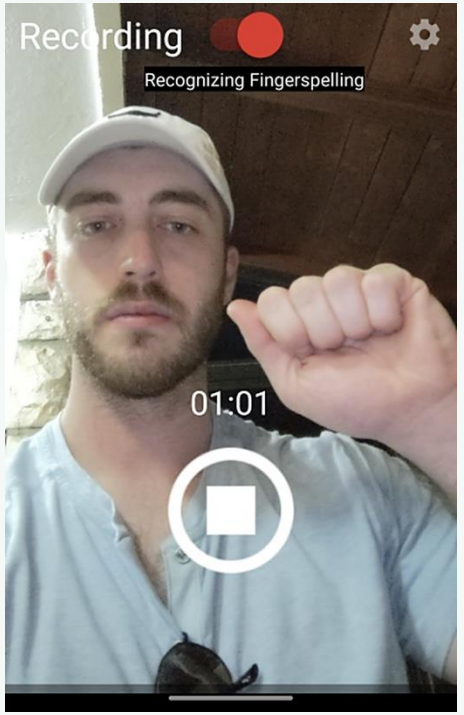
Fingerspelling on a phone camera was faster and more accurate than typing on the keyboard (Tap to Sign).

LARGE-SCALE DATA · CVPR 2025

FSboard: a large fingerspelling dataset recorded by Deaf signers on their own phones, with Google and DPAN.

DEAF-LED

Built with Deaf researchers and the Deaf Professional Arts Network (DPAN), from the questions to the data.



Tap to Sign · MobileHCI 2023



FSboard · CVPR 2025

Sign language for learning

9:25 – 9:40 · 15 min

EXAMPLE 2 **Families and healthcare**

FAMILIES

Hearing parents learning to sign with their Deaf and DeafBlind children. PopSign game and dataset, NeurIPS 2023.

HEALTHCARE

Sign vocabulary for body parts, symptoms, and medical terms. ASSETS 2025.

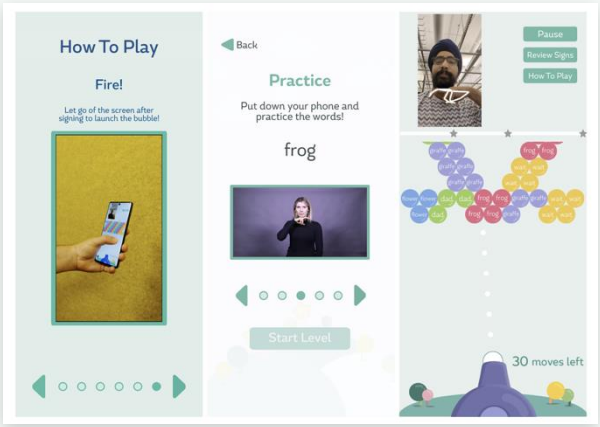
PARTNER

Commission for Deaf & Hard of Hearing

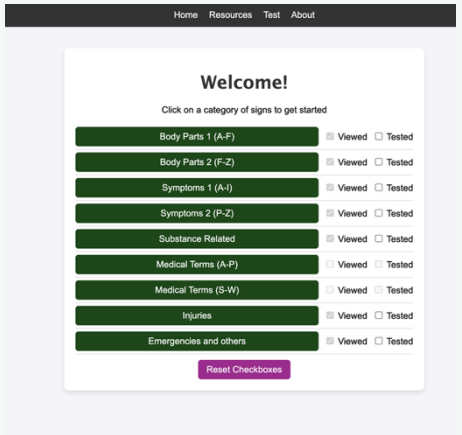
DIDACTIC RECIPROCITY

Designing for people with disabilities, and also for the communities around them: Deaf signers teach, hearing families learn.









Welcome | AI | **Design** | Reflect | Share | Break | Breakouts | Report | Debrief

Captions that carry emotion

9:25 – 9:40 · 15 min

EXAMPLE 3 **Deaf & hard-of-hearing viewers**

THE CHALLENGE

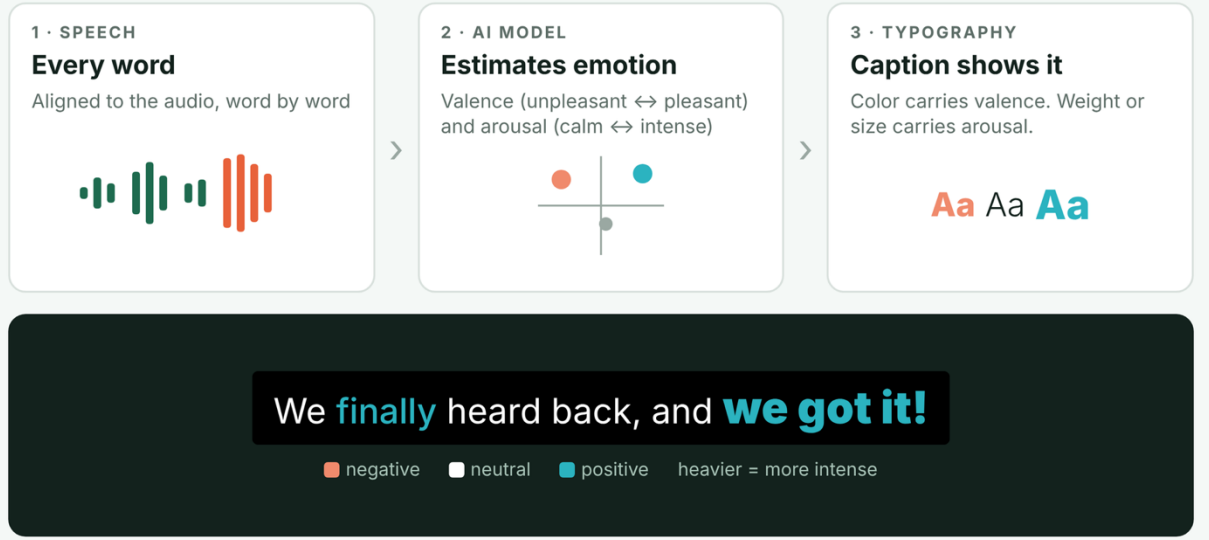
Captions drop tone of voice: how something is said, not just what.

WHAT WE BUILT

Caption Royale: AI estimates emotion from each spoken word, and the caption's typography shows it.

PUBLISHED

CHI 2024 (Honorable Mention), ASSETS 2024, CHI 2025





+ HAPTICS

A wristband adds emotion through vibration. Tactile Emotions, CHI 2025.

Civic AI for accessibility barriers

9:25 – 9:40 · 15 min

EXAMPLE 4 **Civic & disability organizations**

Dashboards and AI that help civic groups, residents, and local governments see and act on accessibility barriers.

> “If I could ask, ‘Give me a list of all ADA sidewalk problems on North State Street between X and Y,’ and it answered accurately and verifiably, that would be game-changing.”

AI summaries with local citations
Answers linked to authoritative records.

Escalate to a human
A visible path to staff when automation falls short.

Mark where AI is used
Plain language and clear onboarding.

NO ONE SIZE FITS ALL

Civic AI has to adapt to different people, roles, and access needs.

NEW NSF AWARD · 2026

Civic AI that connects residents, communities, and local governments.

Your one problem

9:40 – 10:10 · 30 min

Pick one community engagement challenge where AI could help, hurt, create barriers, or lower them.

1	**The problem**	One sentence
2	**Who is affected**	Partners, students, residents
3	**Where AI might help**	Even a little
4	**Where AI worries you**	Risks, harms, trust
5	**Who isn't at the table yet**	Voices to bring in

9:40 · 15 min to write

9:55 · 15 min to share

◯ Circle

the one line you most want to dig into: a hope or a worry.

Share in 1–2 sentences

Your problem, plus what you circled. Sticky note goes on the wall.

10:10 – 10:25 · 15 min

Break

15

minutes

Back at 10:25

While you're away

1. We group your sticky notes into 2–4 shared themes
2. Each theme becomes a breakout starting point
3. Add another sticky note if something new comes up

Breakout groups and share-outs

10:25 – 11:20 · 55 min

Group A
with Nick

Group B
with Saad

Pick the group closest to your problem.

10:25	15 min	**Elaborate the problem**	Add context on sticky notes. What's really going on?
10:40	15 min	**Who else belongs in this conversation?**	Partners and community members you could bring in.
10:55	15 min	**Limits of AI**	Where should human agency stay? Where does human-centered work stand after AI?

11:10 **Share out: 5 min per group. One problem, one partner to bring in, one line AI shouldn't cross.**

Welcome | AI | Design | Reflect | Share | Break | **Breakouts** | **Report** | Debrief

Debrief

11:20 – 11:30 · 10 min

1. One idea you'll take home
2. One open question
3. What follow-up would help you?

saadh.info/pace2

nsmattei@tulane.edu · saadhassan@tulane.edu

Thank you!